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Crop Growth Estimation System Using Machine Vision

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Abstract

According to the philosophy of Precision Farming, the status of crops during their growing stages is important information for crop cultivation tasks and management.
The system which was developed in this research involves the vegetation cover area of plant being determined by the vision system and the image processing technique, and the crop status, i.e., the plant height, the leaf length, and the dry matter, being estimated with the specific functions.
The specific functions which show the relationship between the vegetation cover area of plants and the measured actual plant dimensions were analyzed using a growth curve (the Gompertz curve) and an exponential function. The Gompertz curve was used for the estimation of the dry mass of the plants. For the leaf length and the plant height, the exponential function worked well compared to the growth curve. Based on the results, the crop growing status could be estimated using crop images and calculated equations.

Keywords

Crop status, Exponential function, Growth curve, Map, Precision farming, Regression analysis, Segmentation, Vision system

INTRODUCTION

Information on crop growth during each growing stage is one of the most important indexes of optimum cultivation management. According to the crop growth and the crop health, Precision Farming can determine the number and varieties of fertilizers and chemicals. This leads to a decrease in surplus material. Its benefits include a reduction in cost for farmers and the saving of the environment. In addition, information on the current status of crops is useful for planning the following cultivation schedule.
Remote sensing technology is one of the most suitable sensors for measuring information on the status of crops growing due to the low tasks and the real-time sensing with no damage to the crops. Machine vision is one type of sensor for detecting the crop status [1]. Color images offer much information, however, having an understanding of the information needed from the image is important technology. Research which focused on specific crop status data including the cotton height [2], the chlorophyll contents of corn [3] [4], and the biomass of the crop [5].
The objective of this paper was to develop a crop status estimation system using the vision system and the modeling of crop growth using images.

METHODS

Experimental apparatus

Figure 1 shows the experimental apparatus. A digital color camera (Sony, XC-777A) was placed in front of the sprayer (Iseki, JK-14). The lens height was 2.48 m from the ground surface. The images were recorded using a digital video recorder (Sony, GV-D900). The frame rate of the video recording was 30 frames/s. The recorded images were transferred to images with 640×480 pixels and an RGB of 24 bits using a digital video capture board (Canopus, DVRaptor II).
Since a wide-angle lens with a horizontal angle of 79.18° (Sony, VCL-03S12XM) was used in this system, six rows of crops were captured in each image. The middle four rows were used for the analysis. The traveling velocity of the tractor was set at 0.5 m/s.

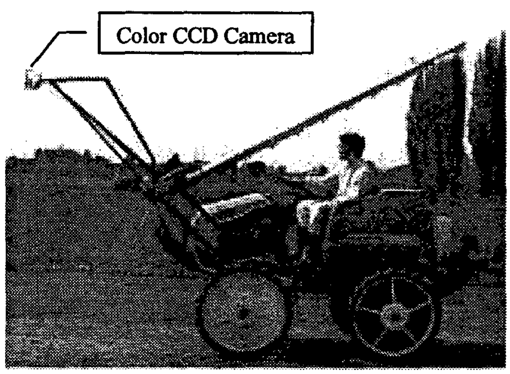

Figure 1 Experimental apparatus.

Tested crops

Soybean and sugar beet crops were tested to capture the images. The distances between the rows were 660 mm for both the soybeans and the sugar beets. The images of both crops were recorded three times under different growing stages from June to September of 2002, at the experimental farm of the Field Science Center at the Northern Biosphere of Hokkaido University. The length of both fields was about 170 m. The widths of the soybean and the sugar beet fields were 10.56 m and 2.64 m, respectively.

In the case of the soybeans, the height was measured by hand every 5 m in the field. They were identified using a maker which was placed beside the target plant. After capturing the images, the plants whose height was measured were removed and their dry matter was measured the total number of samples was thirty plants along one crop row among the captured crop rows.

In the case of the sugar beets, the maximum length of the leaf was measured. Since the leaves of sugar beets grow horizontally, it was considered that the height did not indicate a typical plant size. The measured samples were taken every 5 m, as with the soybeans. Markers were also placed to identify the sampled plants. After capturing the images, the plants whose leaf length was measured, were removed and the dry matter was measured.

Segmentation

Since a wide-angle lens was used, the captured images were warped. Therefore, it was necessary for the images to be corrected. Figure 2 shows a corrected digital color image of the crop rows (soybeans). The latitude and the longitude lengths in the image were 3.40 m and 2.55m, respectively.

Due to the crops being green in color, Tosaka's method [6] was used to segment the crop portion from the soil. This method focuses on the intensity of the RGB in the images. Through the principal component analysis, the green information in the images was enhanced to separate the green plant portion from the background, namely, the soil. Equation (1) explains this feature. Z was named as CIVE (Color Index of Vegetation Extraction) in this paper,

$$Z = 0.441 \times R - 0.811 \times G + 0.385 \times B + 18.78745 \quad (1)$$

where R, G, and B were the values of the RGB intensity of each pixel. Using the CIVE the segmented images of the green plants were easily converted from the color images instead of NIR images.

Figure 3 shows the segmented image of figure 2. A discriminant analysis [7] was used to determine the threshold level.

Figure 2 Digital color image of soybeans.

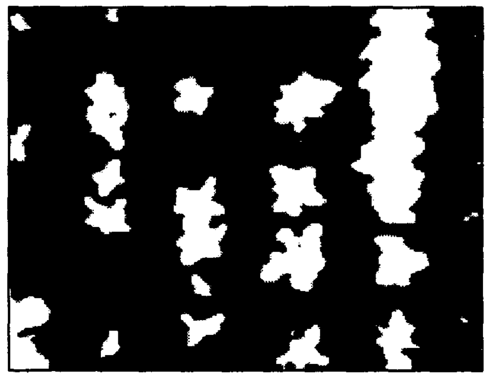

Figure 3 Binary image of figure 2.

Vegetation cover area

The plant dimensions in the images were defined as the vegetation cover area by equation (2), namely,

$$S = \frac{m}{M} \times A \quad (2)$$

in which S was the vegetation cover area of a plant, m was the number of pixels of a plant, M was the number of pixels of the images (= 307 k pixels), and A was the actual ground based area (= 8.67 m^2).

Regression analysis

To evaluate the relationship between the vegetation cover area and the measured actual indexes of the plants (height, dry mass, etc.), a regression analysis was tried. The Gompertz curve (equation (3)), one of the growth curves, and an exponential function were used for the analysis,

$$\frac{dy}{dx} = K(A - y)y \quad (3)$$

where A and k were coefficients. A least-squares curve, using a linear Taylor series [8], was applied to determine these coefficients.

RESULTS AND DISCUSSION

Crop growth status

Tables 1 and 2 show the growth status of the soybean and the sugar beet crops, respectively. The averages are shown for each day.

Table 1 Status of soybean growth.

Date	Height (mm)	Dry mass (g)
6/8	44.7	0.22
6/13	50.9	0.21
6/20	96.3	0.50
6/27	126.6	0.97
7/4	170.6	2.36
8/2	415.6	27.23

Table 2 Status of sugar beet growth.

Date	Leaf length (mm)	Root length (mm)	Dry mass (g)	
			Leaf	Root
6/8	37.4	27.4	0.07	0.04
6/13	61.2	48.6	0.09	0.01
6/20	112.4	64.5	0.60	0.11
6/27	181.0	108.5	3.38	0.36
7/4	260.4	152.9	10.93	3.22
8/2	433.4	213.1	57.18	56.75
9/3	510.3	238.1	114.45	202.16
9/27	508.1	312.5	124.64	335.84

Analysis of crop growth

Figures 4 and 5 show the dry mass and the height of the soybean plants to the vegetation cover area from 5 days to 55 days after the emergence, respectively.

In the case of the soybean dry mass, the index of the correlation was calculated as 0.661. The plotted data was varied, however, and we can see a trend in the growth curve. The Gompertz curve was fitted in equation (4). The R.M.S. was calculated as 8.92 g, namely.

$$y=\frac{69.46\times 10^{1.62\times10^{-3}(x-1280.005)}}{1+10^{1.62\times10^{-3}(x-1280.005)}} \tag{4}$$

In the case of the soybean height, both the Gompertz curve (bold line) and the exponential function (thin line) were calculated as equations (5) and (6). The R.M.S. for each curve was calculated as 3.424 and 5.367, respectively. The indexes of correlation in each equation were calculated as 0945 and 0.858, respectively.

$$y=\frac{54.818\times 10^{1.52\times10^{-3}(x-521.209)}}{1+10^{1.52\times10^{-3}(x-521.209)}} \tag{5}$$

$$y=(1-e^{-0.00314x}) \tag{6}$$

It was deduced that the exponential function for the actual soybean height was a better fit than the Gompertz curve in the earlier stages of the soybean growth. However, in the middle and late stages, the growth curve was able to evaluate the height well.

Figures 6 to 9 show the results for the sugar beet crop. The dry mass of leaves and roots were analyzed only by the Gompertz curve. On the other hand, the maximum leaf length and root length were analyzed by both the Gompertz curve (bold line) and the exponential function (thin line).

Equations (7) (8), (9) and (10) represent the Gompertz curve for the dry mass of the leaves, the dry mass of roots, the maximum leaf length and the root length, respectively.

$$y=\frac{164.184\times 10^{4.564\times10^{-4}(x-4273)}}{1+10^{4.564\times10^{-4}(x-4273)}} \tag{7}$$

$$y=\frac{346.776\times 10^{6.989\times10^{-4}(x-4292.873)}}{1+10^{6.989\times10^{-4}(x-4292.873)}} \tag{8}$$

$$y=\frac{101.102\times 10^{1.59\times10^{-4}(x-4900.492)}}{1+10^{1.59\times10^{-4}(x-4900.492)}} \tag{9}$$

$$y=\frac{97.935\times 10^{1.13\times10^{-4}(x-9476.933)}}{1+10^{1.13\times10^{-4}(x-9476.933)}} \tag{10}$$

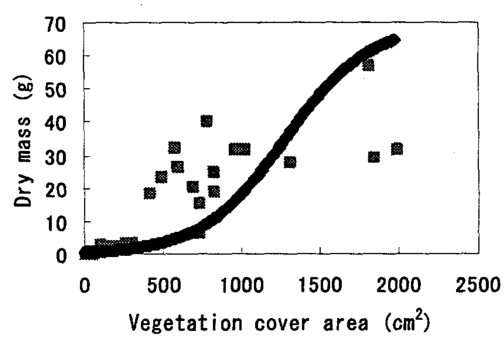

Figure 4 Dry mass of soybean to vegetation cover area.

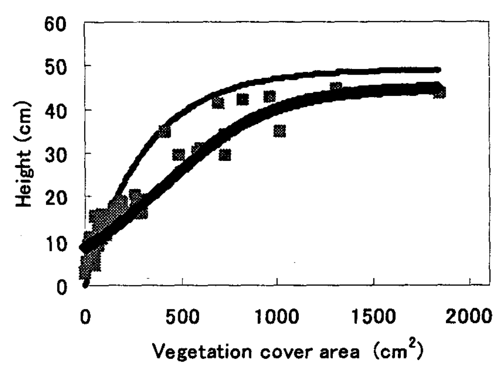

Figure 5 Soybean height to vegetation cover area.

Equations (11) and (12) represent the exponential function of the maximum leaf length and root length, respectively.

$$y=51.6(1-e^{-0.000871x}) \tag{11}$$

$$y=27.3(1-e^{-0.00117x}) \tag{12}$$

Table 3 shows the R.M.S. and the index of correlation for these four measured subjects.

As in the case of the soybean crop, the plotted data for the dry mass of the leaves and the roots varied. However, there seems to be a trend in the growth curve. The indexes of correlation for both were relatively high, but the R.M.S. values were too high to estimate the dry mass of the plants from the vegetation cover area in the images.

As for the maximum leaf length and root length, the exponential functions fit well compared to the Gompertz curve. The values of the R.M.S. were smaller and the index of correlation worked well compared to the Gompertz curve. The major reason why the exponential function worked well is that it did not use collected data on the leaf and the root length from the earlier growing stages. In the earlier growing stages, the plant size was too small to detect with the vision system. This depended on the resolution of the vision system.

It was summarized that the crop growth status could be estimated using the images and the calculated equations instead of the actual hand measurements.

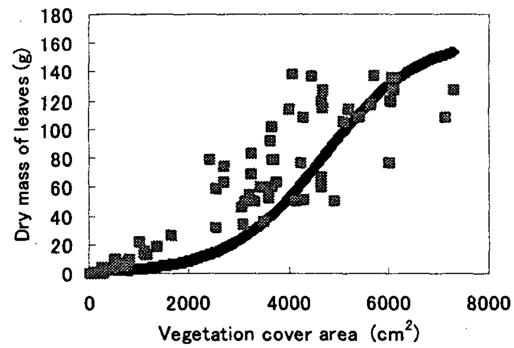

Figure 6 Dry mass of sugar beet leaves to vegetation cover area.

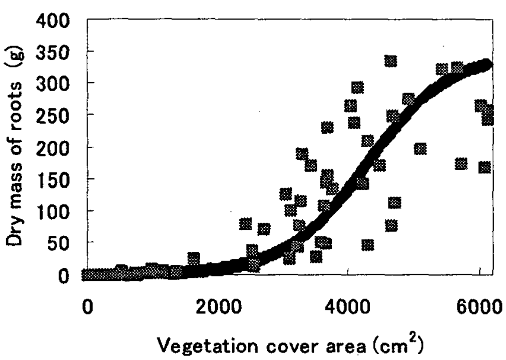

Figure 7 Dry mass of sugar beet roots to vegetation cover area.

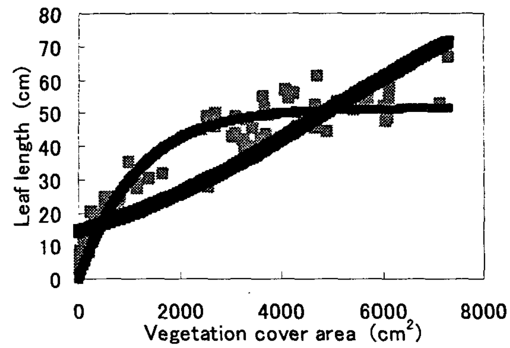

Figure 8 Maximum leaf length of sugar beet to vegetation cover area.

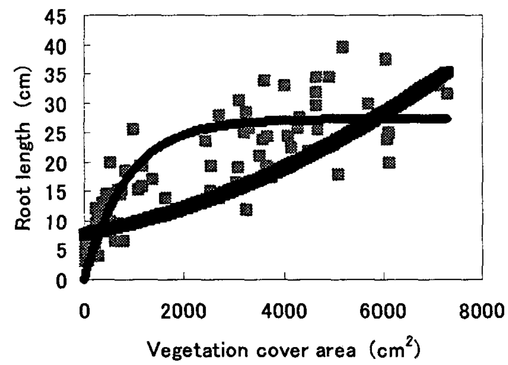

Figure 9 Root length of sugar beet to vegetation cover area.

Table 3 R.M.S. and index of correlation for dry mass of roots, dry mass of leaves, maximum leaf length and root length of sugar beets.

	Gompertz		Exponential function	
	R.M.S.	Index of correlation	R.M.S.	Index of correlation
Dry mass of leaves	51.59	0.804	-	-
Dry mass of roots	26.07	0.831	-	-
Leaf length	7.99	0.895	5.13	0.958
Root length	6.25	0.770	5.27	0.841

CONCLUSION

The crop status was estimated using the vision system. The index was the vegetation cover area. To evaluate the crop growing status, the relationship between the vegetation cover area and the actual plant status was evaluated by a growth curve (the Gompertz curve) and an exponential function. The Gompertz curve could be used for the estimation of the dry mass of plants. As for the leaf length and the plant height, the exponential function worked well compared to the growth curve.

Based on these results, the crop growing status can be estimated using crop images and calculated equations.
The accuracy of the plant status estimation, using the analyzed function, was not acceptable for every plant status. However, the aim of this project was to estimate the plant status in a specific area, not every plant. With regard to this endeavor, the system introduced to this project was found to be acceptable.

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